

MOST ASKED

SQL

INTERVIEW QUESTIONS

at MAANG Companies





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Everyone learns uniquely.

Master SQL concepts by through these important interview questions.

Take the help of this doc, and crack your SQL questions like a pro.

Medium-Level SQL Questions

Q1.

Explain the difference between **INNER JOIN** and **LEFT JOIN**.

Ans.

An '**INNER JOIN**' returns only the rows where there is a match in both tables. A '**LEFT JOIN**' returns all rows from the left table and matched rows from the right table, and '**NULL**' for non-matching rows in the right table.

Q2.

What is the purpose of the **GROUP BY** clause?

Ans.

The **GROUP BY** clause groups rows that have the same values in specified columns into summary rows, like "find the number of customers in each country."

Q3.

How can you delete duplicate rows in a SQL table?

Ans.

sql

Copy code

```
DELETE FROM your_table
WHERE id NOT IN (
    SELECT MIN(id)
    FROM your_table
    GROUP BY column_with_duplicates
);
```

Q4.

What are subqueries and how are they used?

Ans.

Subqueries are nested queries used within a main query to return data that will be used in the main query as a condition to further restrict the data to be retrieved.

Q5.

Explain the concept of a 'VIEW' in SQL.

Ans.

A 'VIEW' is a virtual table based on the result set of an SQL query. It can simplify complex queries, improve security by restricting data access, and present a consistent, logical view of the data.

Q6.

What is a 'UNION' operator?

Ans.

The 'UNION' operator is used to combine the result sets of two or more 'SELECT' statements. Each 'SELECT' statement must have the same number of columns in the result sets with similar data types.

Q7.

How do you use the 'CASE' statement in SQL?

Ans.

sql

Copy code

```
SELECT name,  
CASE  
  WHEN score >= 90 THEN 'A'  
  WHEN score >= 80 THEN 'B'  
  ELSE 'C'  
END as grade  
FROM students;
```

Q8.

What is an 'INDEX' and why is it important?

Ans.

An index is a database object that improves the speed of data retrieval operations on a table at the cost of additional writes and storage space.

Q9.

Describe the use of 'TRIGGERS' in SQL.

Ans.

Triggers are database objects that automatically execute a specified SQL procedure when a certain event occurs on a table or view, such as an 'INSERT', 'UPDATE', or 'DELETE'.

Q10.

Explain the concept of 'NORMALIZATION'.

Ans.

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity. It involves dividing large tables into smaller ones and defining relationships between them.

Advanced SQL Questions

Q11.

What is a 'CTE' (Common Table Expression)?

Ans.

sql

Copy code

```
WITH Sales_CTE (SalesPerson, TotalSales) AS (  
    SELECT SalesPerson, SUM(SalesAmount)  
    FROM Sales  
    GROUP BY SalesPerson  
)  
SELECT * FROM Sales_CTE;
```


Q12.

Explain the differences between 'DELETE', 'TRUNCATE', and 'DROP' commands.

Ans.

'DELETE': Removes rows one at a time and logs individual row deletions. It can include a WHERE clause.

'TRUNCATE': Removes all rows from a table without logging individual row deletions. Cannot include a 'WHERE' clause.

'DROP': Removes a table from the database entirely, including its structure.

Q13.

How do you optimize a slow-running query?

Ans.

Techniques include indexing, query rewriting, reducing the number of joins, avoiding SELECT *, and analyzing the execution plan.

Q14.

What are 'window functions' in SQL?

Ans.

```
sql Copy code  
  
SELECT name, department,  
       salary,  
       RANK() OVER (PARTITION BY department ORDER BY salary DESC) AS rank  
FROM employees;
```

Q15.

Discuss the concept of 'ACID' properties in transactions.

Ans.

Atomicity: Ensures that all operations within a transaction are completed; if not, the transaction is aborted.

Consistency: Ensures that a transaction brings the database from one valid state to another.

Isolation: Ensures that transactions are securely and independently processed at the same time without interference.

Durability: Ensures that once a transaction is committed, it will remain so, even in the event of a power loss, crash, or error.

Q16.

How do you handle transactions in SQL?

Ans.

sql

 Copy code

```
BEGIN TRANSACTION;  
-- SQL statements  
COMMIT;
```

Q17.

What is the difference between 'HAVING' and 'WHERE' clauses?

Ans.

'WHERE' is used to filter records before any groupings are made, while 'HAVING' is used to filter values after grouping.

Q18.

Explain the concept of 'sharding' in databases.

Ans.

Sharding is a type of database partitioning that splits large databases into smaller, more manageable pieces, called shards, which can be spread across multiple servers.

Q19.

What are 'stored procedures' and how are they used?

Ans.

Stored procedures are precompiled collections of SQL statements stored under a name and processed as a unit. They can accept parameters and are used to perform tasks like data validation or access control.

Q20.

Explain the concept of 'Data Warehousing'.

Ans.

Data warehousing involves the collection, storage, and management of large volumes of data from multiple sources to provide meaningful business insights. It typically involves ETL processes and uses schemas like star or snowflake schema.

Scenario-Based SQL Questions

Scenario 1:

Employee Management System

Tables:

- Employees

sql

Copy code

```
CREATE TABLE Employees (  
  EmployeeID INT PRIMARY KEY,  
  FirstName VARCHAR(50),  
  LastName VARCHAR(50),  
  DepartmentID INT,  
  Salary DECIMAL(10, 2),  
  HireDate DATE  
);
```

- Departments

sql

Copy code

```
CREATE TABLE Departments (  
  DepartmentID INT PRIMARY KEY,  
  DepartmentName VARCHAR(50)  
);
```


Questions:

Q21.

Find the second highest salary in the Employees table.

Ans.

sql

 Copy code

```
SELECT MAX(Salary) AS SecondHighestSalary
FROM Employees
WHERE Salary < (SELECT MAX(Salary) FROM Employees);
```

Q22.

List all employees who have been hired in the last year.

Ans.

sql

 Copy code

```
SELECT * FROM Employees
WHERE HireDate >= DATEADD(year, -1, GETDATE());
```


Q23.

Find the average salary for each department.

Ans.

sql

 Copy code

```
SELECT d.DepartmentName, AVG(e.Salary) AS AverageSalary
FROM Employees e
JOIN Departments d ON e.DepartmentID = d.DepartmentID
GROUP BY d.DepartmentName;
```

Q24.

List all departments along with the count of employees in each department.

Ans.

sql

 Copy code

```
SELECT d.DepartmentName, COUNT(e.EmployeeID) AS EmployeeCount
FROM Departments d
LEFT JOIN Employees e ON d.DepartmentID = e.DepartmentID
GROUP BY d.DepartmentName;
```

Scenario 2:

Online Retail Database

Tables:

- Orders

sql

 Copy code

```
CREATE TABLE Orders (  
  OrderID INT PRIMARY KEY,  
  CustomerID INT,  
  OrderDate DATE,  
  TotalAmount DECIMAL(10, 2)  
);
```

- OrderDetails

sql

 Copy code

```
CREATE TABLE OrderDetails (  
  OrderDetailID INT PRIMARY KEY,  
  OrderID INT,  
  ProductID INT,  
  Quantity INT,  
  Price DECIMAL(10, 2)  
);
```

- Products

sql

 Copy code

```
CREATE TABLE Products (  
    ProductID INT PRIMARY KEY,  
    ProductName VARCHAR(100),  
    CategoryID INT,  
    Price DECIMAL(10, 2)  
);
```

Questions:

Q25.

Find the top 3 most sold products in the last month.

Ans.

sql

 Copy code

```
SELECT TOP 3 p.ProductName, SUM(od.Quantity) AS TotalSold
FROM OrderDetails od
JOIN Products p ON od.ProductID = p.ProductID
JOIN Orders o ON od.OrderID = o.OrderID
WHERE o.OrderDate >= DATEADD(month, -1, GETDATE())
GROUP BY p.ProductName
ORDER BY TotalSold DESC;
```

Q26.

Calculate the total sales for each product category.

Ans.

sql

 Copy code

```
SELECT c.CategoryName, SUM(od.Quantity * od.Price) AS TotalSales
FROM OrderDetails od
JOIN Products p ON od.ProductID = p.ProductID
JOIN Categories c ON p.CategoryID = c.CategoryID
GROUP BY c.CategoryName;
```

Q27.

Find all customers who placed orders in the last 7 days.

Ans.

sql

 Copy code

```
SELECT DISTINCT CustomerID
FROM Orders
WHERE OrderDate >= DATEADD(day, -7, GETDATE());
```

Q28.

Calculate the average order value for each customer.

Ans.

sql

 Copy code

```
SELECT CustomerID, AVG(TotalAmount) AS AverageOrderValue
FROM Orders
GROUP BY CustomerID;
```

Q29.

Identify the product that generated the highest revenue.

Ans.

sql

 Copy code

```
SELECT p.ProductName, SUM(od.Quantity * od.Price) AS TotalRevenue
FROM OrderDetails od
JOIN Products p ON od.ProductID = p.ProductID
GROUP BY p.ProductName
ORDER BY TotalRevenue DESC
LIMIT 1;
```

Q30.

List customers who have placed more than 5 orders.

Ans.

sql

Copy code

```
SELECT CustomerID, COUNT(OrderID) AS OrderCount
FROM Orders
GROUP BY CustomerID
HAVING COUNT(OrderID) > 5;
```

Q31.

Find the month-over-month growth rate of total sales.

Ans.

sql

Copy code

```
SELECT
    DATE_FORMAT(OrderDate, '%Y-%m') AS Month,
    SUM(TotalAmount) AS TotalSales,
    (SUM(TotalAmount) - LAG(SUM(TotalAmount), 1) OVER (ORDER BY DATE_FORMAT
    (OrderDate, '%Y-%m')) / LAG(SUM(TotalAmount), 1) OVER (ORDER BY DATE_FORMAT
    (OrderDate, '%Y-%m')) * 100 AS GrowthRate
FROM Orders
GROUP BY DATE_FORMAT(OrderDate, '%Y-%m');
```


Q32.

Identify the employees who processed the highest number of orders.

Ans.

sql

Copy code

```
SELECT e.EmployeeID, COUNT(o.OrderID) AS OrderCount
FROM Orders o
JOIN Employees e ON o.EmployeeID = e.EmployeeID
GROUP BY e.EmployeeID
ORDER BY OrderCount DESC
LIMIT 1;
```

Q33.

Find the products that have not been sold in the last 6 months.

Ans.

sql

Copy code

```
SELECT p.ProductName
FROM Products p
LEFT JOIN OrderDetails od ON p.ProductID = od.ProductID
LEFT JOIN Orders o ON od.OrderID = o.OrderID AND o.OrderDate >= DATEADD
(month, -6, GETDATE())
WHERE o.OrderID IS NULL;
```

Q34.

List customers and the total amount they have spent.

Ans.

sql

 Copy code

```
SELECT CustomerID, SUM(TotalAmount) AS TotalSpent
FROM Orders
GROUP BY CustomerID;
```

Q35.

Calculate the average number of items per order.

Ans.

sql

 Copy code

```
SELECT AVG(ItemCount) AS AvgItemsPerOrder
FROM (
    SELECT OrderID, SUM(Quantity) AS ItemCount
    FROM OrderDetails
    GROUP BY OrderID
) AS OrderItemCounts;
```

Q36.

Find overlapping date ranges for product availability.

Ans.

sql

 Copy code

```
CREATE TABLE ProductAvailability (  
    ProductID INT,  
    StartDate DATE,  
    EndDate DATE  
);  
  
SELECT a.ProductID, a.StartDate, a.EndDate, b.StartDate, b.EndDate  
FROM ProductAvailability a, ProductAvailability b  
WHERE a.ProductID = b.ProductID  
AND a.StartDate < b.EndDate  
AND b.StartDate < a.EndDate  
AND a.StartDate <> b.StartDate;
```

Q37.

Retrieve the most recent order for each customer.

Ans.

sql

 Copy code

```
SELECT CustomerID, MAX(OrderDate) AS MostRecentOrderDate
FROM Orders
GROUP BY CustomerID;
```

Q38.

List products that have been ordered more than 100 times.

Ans.

sql

 Copy code

```
SELECT p.ProductName, SUM(od.Quantity) AS TotalOrdered
FROM Products p
JOIN OrderDetails od ON p.ProductID = od.ProductID
GROUP BY p.ProductName
HAVING SUM(od.Quantity) > 100;
```


Q39.

Create a query to find the longest consecutive sequence of orders for each customer.

Ans.

sql

 Copy code

```
WITH ConsecutiveOrders AS (  
    SELECT  
        CustomerID,  
        OrderDate,  
        ROW_NUMBER() OVER (PARTITION BY CustomerID ORDER BY OrderDate) AS RowNum  
    FROM Orders  
)  
SELECT CustomerID, COUNT(*) AS LongestStreak  
FROM ConsecutiveOrders  
GROUP BY CustomerID, DATE_SUB(OrderDate, INTERVAL RowNum DAY)  
ORDER BY LongestStreak DESC;
```

Q40.

Fetch the last N records in a table, ordered by a specific column.

Ans.

sql

 Copy code

```
SELECT *  
FROM Orders  
ORDER BY OrderDate DESC  
LIMIT N;
```


Q41.

Audit data changes in a SQL database.

Ans.

sql

Copy code

```
CREATE TABLE AuditLog (  
    AuditID INT PRIMARY KEY,  
    TableName VARCHAR(50),  
    Action VARCHAR(10),  
    OldValue VARCHAR(255),  
    NewValue VARCHAR(255),  
    ChangeDate DATETIME,  
    ChangedBy VARCHAR(50)  
);  
  
CREATE TRIGGER trgAudit  
AFTER UPDATE ON Orders  
FOR EACH ROW  
BEGIN  
    INSERT INTO AuditLog (TableName, Action, OldValue, NewValue, ChangeDate,   
                          ChangedBy)  
    VALUES ('Orders', 'UPDATE', OLD.TotalAmount, NEW.TotalAmount, NOW(), USER());  
END;
```

Q42.

Design a query to retrieve hierarchical data in a parent-child relationship.

Ans.

sql

Copy code

```
CREATE TABLE Employees (  
    EmployeeID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    ManagerID INT  
);  
  
WITH EmployeeHierarchy AS (  
    SELECT EmployeeID, Name, ManagerID, 1 AS Level  
    FROM Employees  
    WHERE ManagerID IS NULL  
    UNION ALL  
    SELECT e.EmployeeID, e.Name, e.ManagerID, eh.Level + 1  
    FROM Employees e  
    JOIN EmployeeHierarchy eh ON e.ManagerID = eh.EmployeeID  
)  
SELECT * FROM EmployeeHierarchy  
ORDER BY Level, ManagerID;
```

Q43.

Implement soft deletes in SQL.

Ans.

sql

Copy code

```
ALTER TABLE Orders ADD COLUMN IsDeleted BOOLEAN DEFAULT FALSE;

-- Mark as deleted
UPDATE Orders SET IsDeleted = TRUE WHERE OrderID = 1;

-- Retrieve non-deleted orders
SELECT * FROM Orders WHERE IsDeleted = FALSE;
```

Q44.

Write a query to calculate the year-over-year growth of sales.

Ans.

sql

Copy code

```
SELECT
    YEAR(OrderDate) AS Year,
    SUM(TotalAmount) AS TotalSales,
    (SUM(TotalAmount) - LAG(SUM(TotalAmount), 1) OVER (ORDER BY YEAR(OrderDate)))
    / LAG(SUM(TotalAmount), 1) OVER (ORDER BY YEAR(OrderDate)) * 100 AS YoYGrowth
FROM Orders
GROUP BY YEAR(OrderDate);
```


Q45.

Handle data migration from one database to another.

Ans.

sql

 Copy code

```
-- Using MySQL Workbench or similar tools for data migration.  
-- Example:  
mysqldump -u source_user -p source_db > source_db.sql  
mysql -u target_user -p target_db < source_db.sql
```

Q46.

How to implement full-text search in SQL?

Ans.

sql

 Copy code

```
CREATE FULLTEXT INDEX idx_productname ON Products(ProductName);  
  
SELECT * FROM Products  
WHERE MATCH(ProductName) AGAINST('search term');
```

Q47.

Write a query to find the median value in a numeric column.

Ans.

```
sql Copy code

SELECT AVG(value) AS MedianValue
FROM (
    SELECT value
    FROM table
    ORDER BY value
    LIMIT 2 - (SELECT COUNT(*) FROM table) % 2
    OFFSET (SELECT (COUNT(*) - 1) / 2 FROM table)
) AS MedianValues;
```

Q48.

How to perform a bulk insert in SQL?

Ans.

```
sql Copy code

INSERT INTO Orders (OrderID, CustomerID, OrderDate, TotalAmount)
VALUES
(1, 1, '2023-01-01', 100.00),
(2, 2, '2023-01-02', 150.00),
(3, 1, '2023-01-03', 200.00);
```


Q49.

Calculate the running total of sales for each day.

Ans.

sql

Copy code

```
SELECT OrderDate,  
       TotalAmount,  
       SUM(TotalAmount) OVER (ORDER BY OrderDate) AS RunningTotal  
FROM Orders;
```

Q48.

Identify customers who have not placed any orders in the last 6 months.

Ans.

sql

Copy code

```
SELECT CustomerID  
FROM Customers  
WHERE CustomerID NOT IN (  
    SELECT DISTINCT CustomerID  
    FROM Orders  
    WHERE OrderDate >= DATEADD(month, -6, GETDATE())  
);
```



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